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MIL-M-28787/462
10 May 1991

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

SEMICONDUCTOR MEMORY ARRAY, DYNAMIC RAM, 1M MODULE,

KEY CODE JKJ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic semiconductor memory array, dynamic ram, 1M module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5963

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 33.722 failures per million operating hours for class I or 147.912 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be JKJ.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
A2-A16	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
B1-B9	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Power supply stress	IV	S	----------Module shall pass all remaining tests----------						
D1-D10	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D11	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D12	Ibd	V	S	0 0	0.1	0.0	0.1	0.0	0 1	mA
D13	Ibd	V	S	0 0	0.4	0 0	0.4	0.0	0 4	mA
D14	Ibd	V	S	0.0	0.5	0.0	0.5	0.0	0.5	mA
D15-D24	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0 1	mA
D25	Ibd	V	S	-0.03	0.03	-0.03	0.03	-0.03	0.03	mA
D26-D46	Ibd	V	S	0 0	0.1	0.0	0.1	0.0	0 1	mA
E1-E22	Ibe	VI	S	-350.0	-60.0	-350.0	-60.0	-350.0	-60.0	mA
F1-F4	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F5-F8	Iih	VII	X	-0.015	0 015	-0.015	0.015	-0.015	0.015	mA
F9-F14	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F15-F18	Iih	VII	X	-0.015	0 015	-0.015	0.015	-0.015	0.015	mA
F19	Iih	VII	X	-4 8	0.06	-4.8	0.06	-4.8	0.06	mA
F20	Iih	VII	X	-1.6	0.02	-1.6	0.02	-1.6	0.02	mA
F21	Iih	VII	X	-6 4	0 08	-6.4	0.08	-6.4	0.08	mA
F22	Iih	VII	X	-8 0	0 1	-8.0	0.1	-8.0	0.1	mA
F23-F25	Iih	VII	X	-0.015	0.015	-0.015	0.015	-0.015	0.015	mA
F26-F29	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F30-F33	Iih	VII	X	-0.015	0.015	-0.015	0.015	-0.015	0.015	mA
F34-F39	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F40-F43	Iih	VII	X	-0.015	0 015	-0.015	0.015	-0.015	0.015	mA
F44	Iih	VII	X	-0.03	0.03	-0.03	0.03	-0.03	0.03	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F45	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F46-F48	Iih	VII	X	-0.015	0.015	-0.015	0.015	-0.015	0.015	mA
F49-F68	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G1-G4	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G5-G8	Iil	VIII	X	-0.015	0.015	-0.015	0.015	-0.015	0.015	mA
G9-G14	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G15-G18	Iil	VIII	X	-0.015	0.015	-0.015	0.015	-0.015	0.015	mA
G19	Iil	VIII	X	-4.8	0.06	-4.8	0.06	-4.8	0.06	mA
G20	Iil	VIII	X	-1.6	0.02	-1.6	0.02	-1.6	0.02	mA
G21	Iil	VIII	X	-6.4	0.08	-6.4	0.08	-6.4	0.08	mA
G22	Iil	VIII	X	-8.0	0.1	-8.0	0.1	-8.0	0.1	mA
G23-G25	Iil	VIII	X	-0.015	0.015	-0.015	0.015	-0.015	0.015	mA
G26-G29	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G30-G33	Iil	VIII	X	-0.015	0.015	-0.015	0.015	-0.015	0.015	mA
G34-G39	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G40-G43	Iil	VIII	X	-0.015	0.015	-0.015	0.015	-0.015	0.015	mA
G44	Iil	VIII	X	-0.03	0.03	-0.03	0.03	-0.03	0.03	mA
G45	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G46-G48	Iil	VIII	X	-0.015	0.015	-0.015	0.015	-0.015	0.015	mA
G49-G68	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
H1-H24	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J28	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K1-K4	Iozh	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
K5-K8	Iozl	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
L1-L45	Vio	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
M	PBFC	XIII	S	7.57	14.07	7.57	14.07	7.57	14.07	uF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
N	Icc	XIV	X	190	460	190	460	190	460	mA
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	Q	See table XVI						
R1-R12	C1/D	XVIII	Q	2	50	2	50	2	50	pF
R13,R14	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R15-R34	C1/D	XVIII	Q	2	50	2	50	2	50	pF
R35	C1/D	XVIII	Q	2	175	2	175	2	175	pF
R36	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R37-R46	C1/D	XVIII	Q	2	50	2	50	2	50	pF
R47-R68	C1/D	XVIII	Q	4	50	4	50	4	50	pF
T	PSTR	XIX	Q	0	400	0	400	0	400	mV

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/462-1 for class I and M28787/462-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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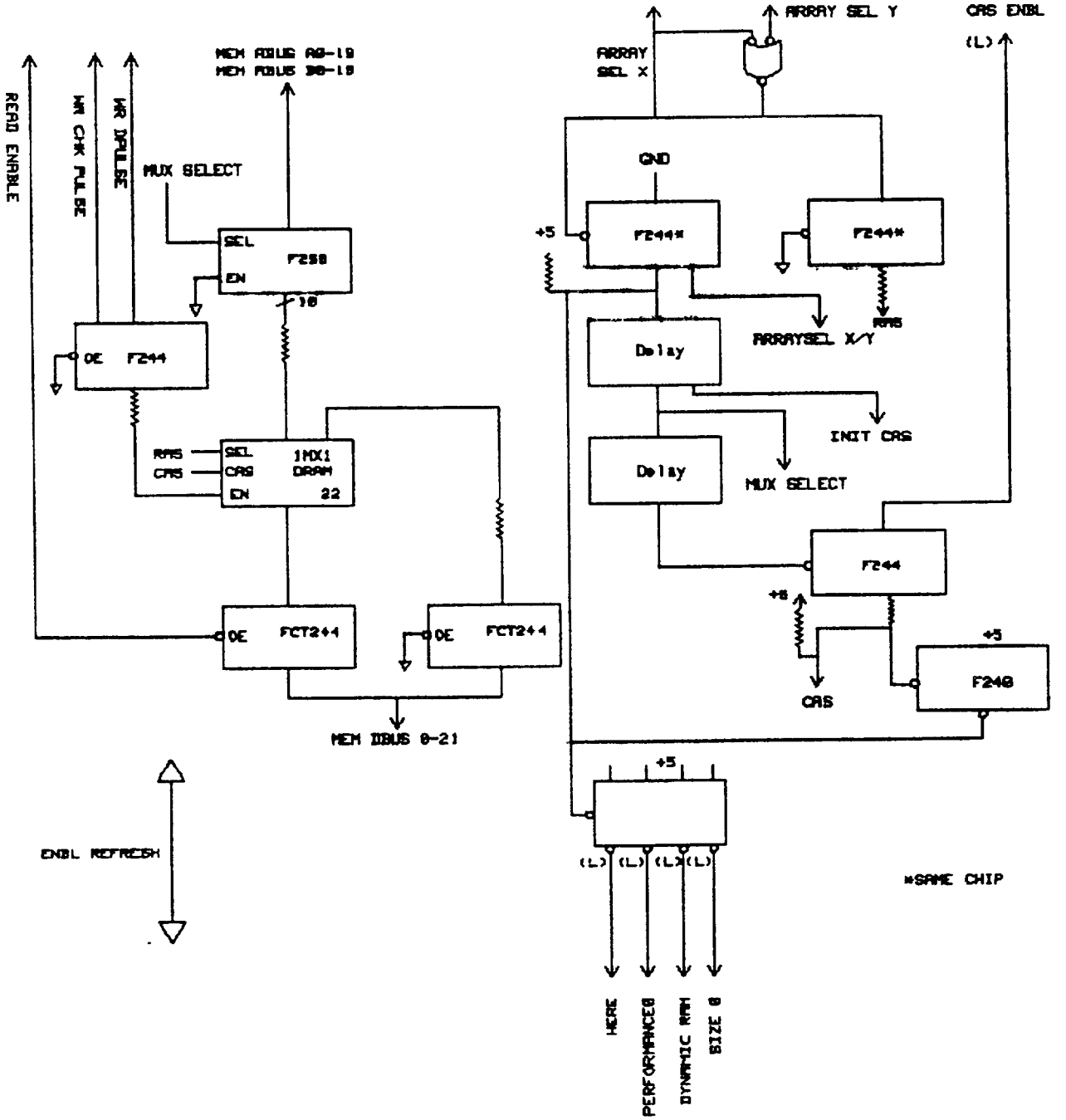


FIGURE 1. JKJ functional block diagram.

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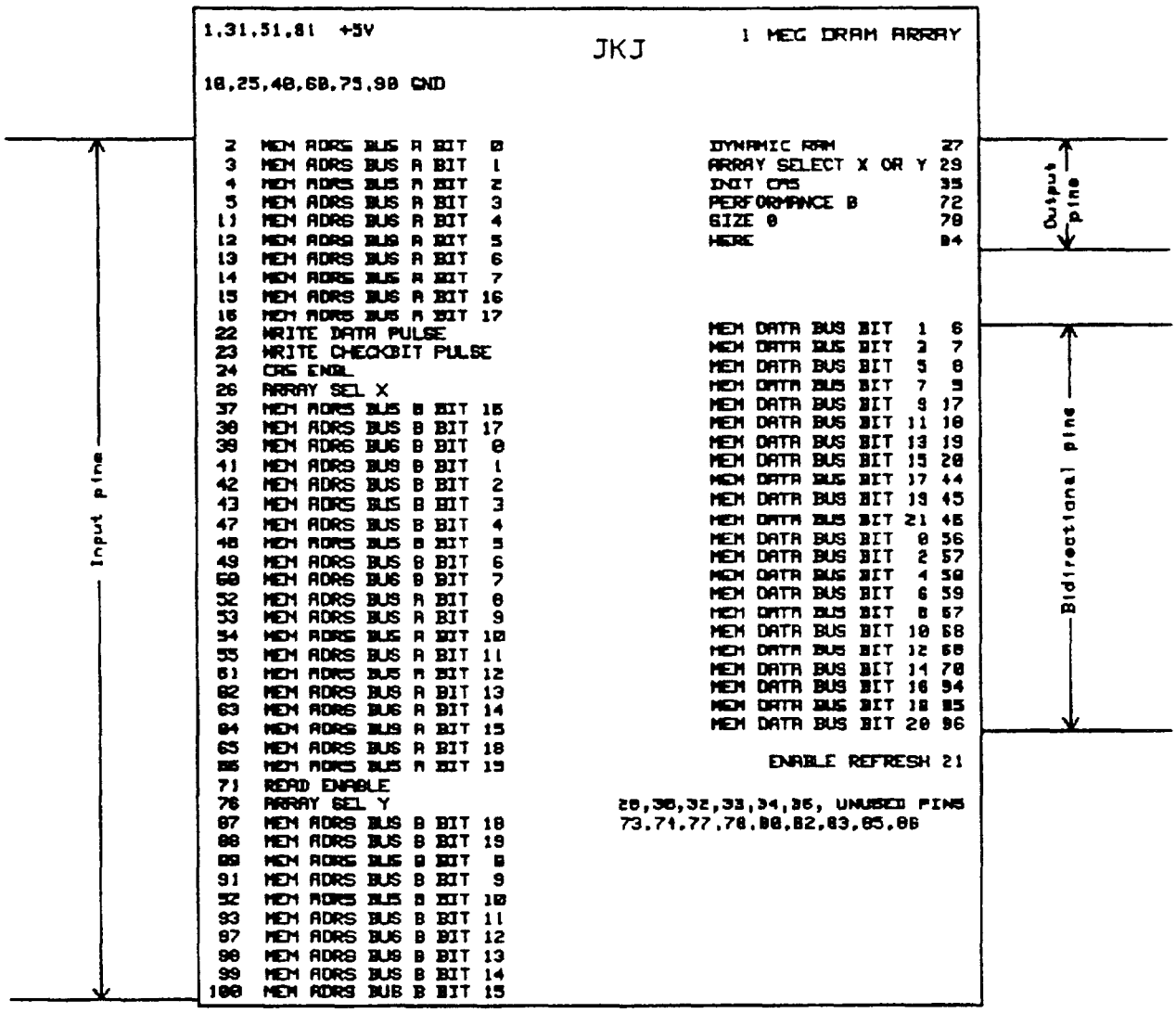


FIGURE 2. JKJ pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

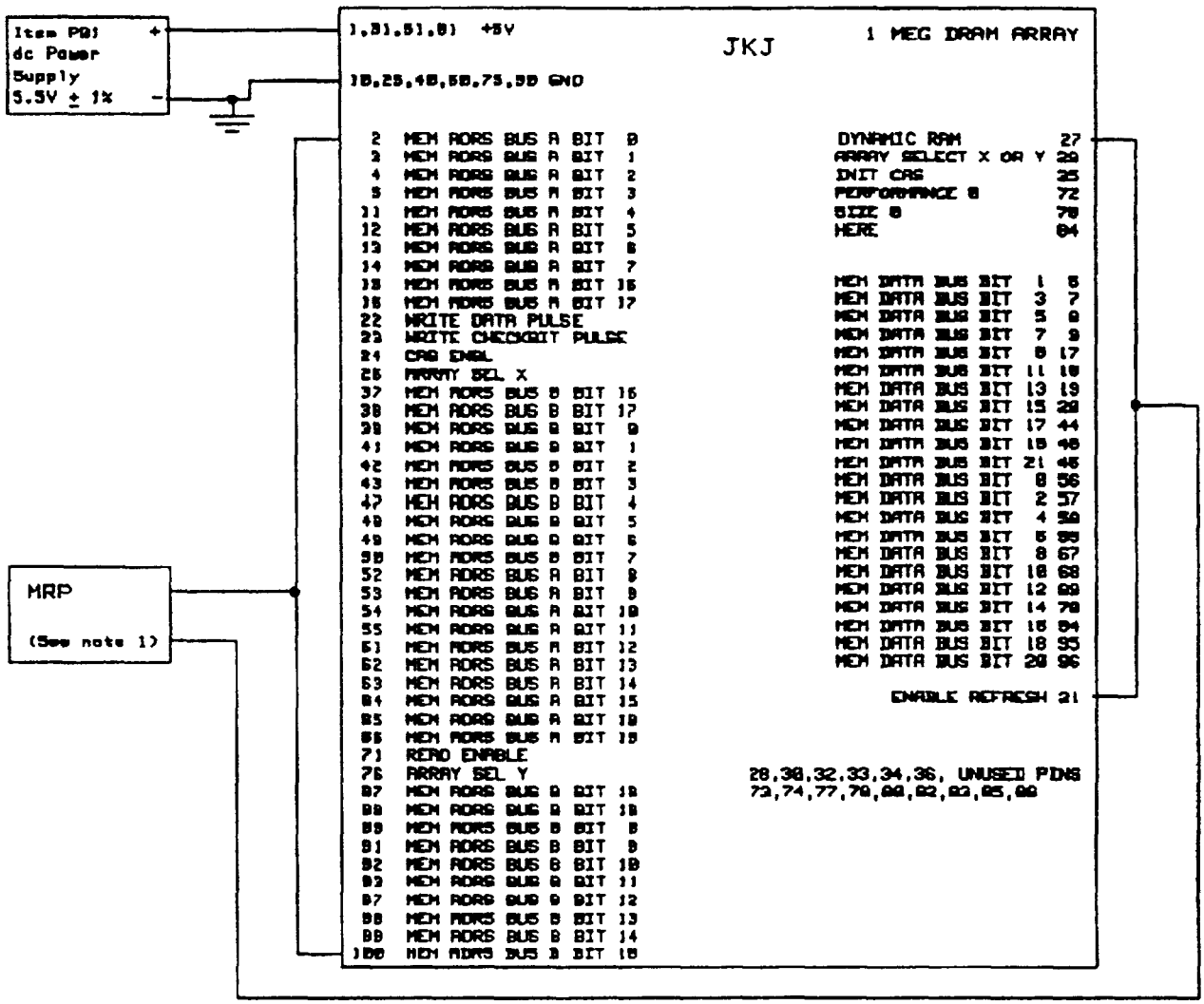
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. JKJ operating temperature and component temperature test circuit.

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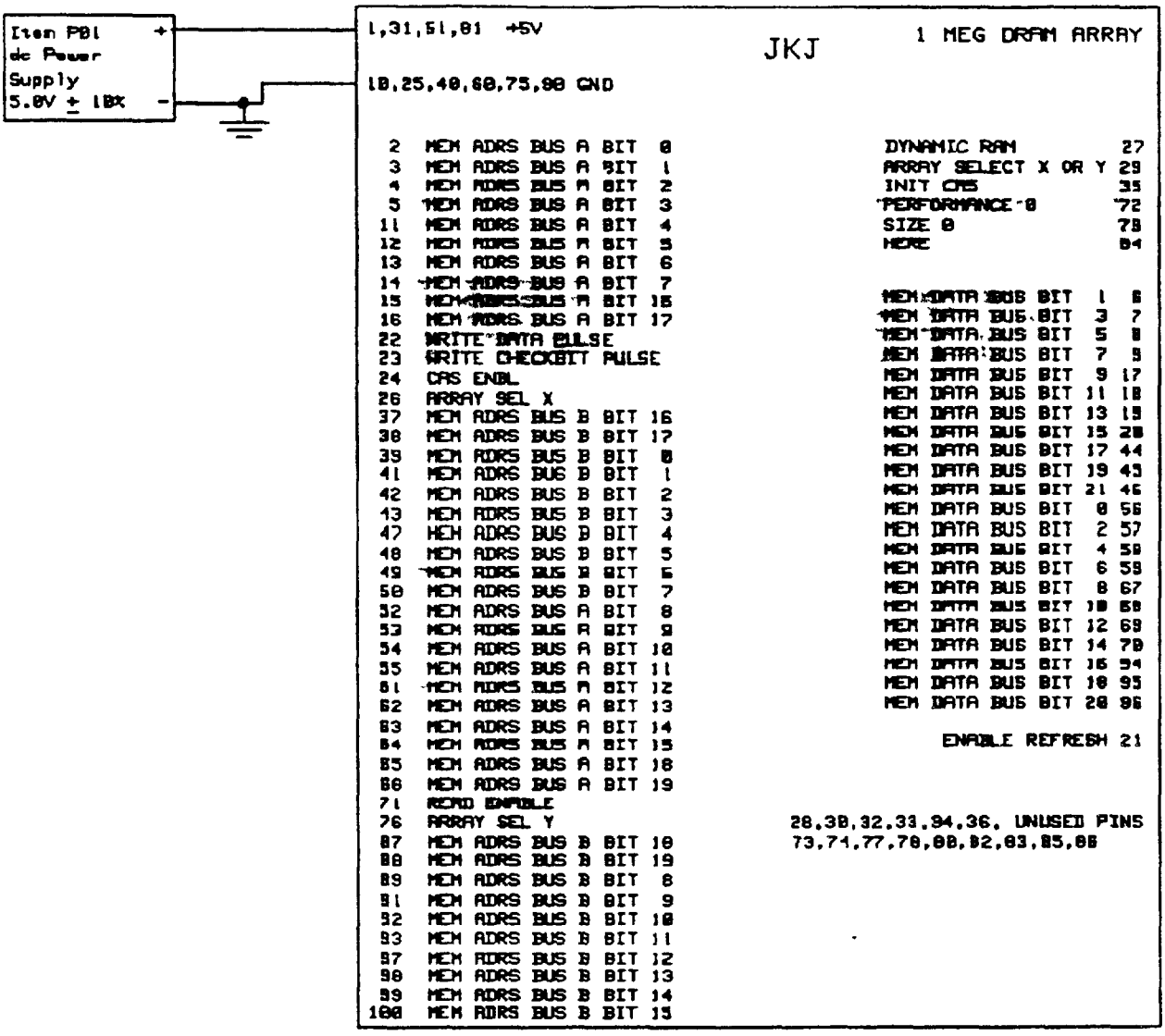


FIGURE 4. JKJ vibration test circuit.

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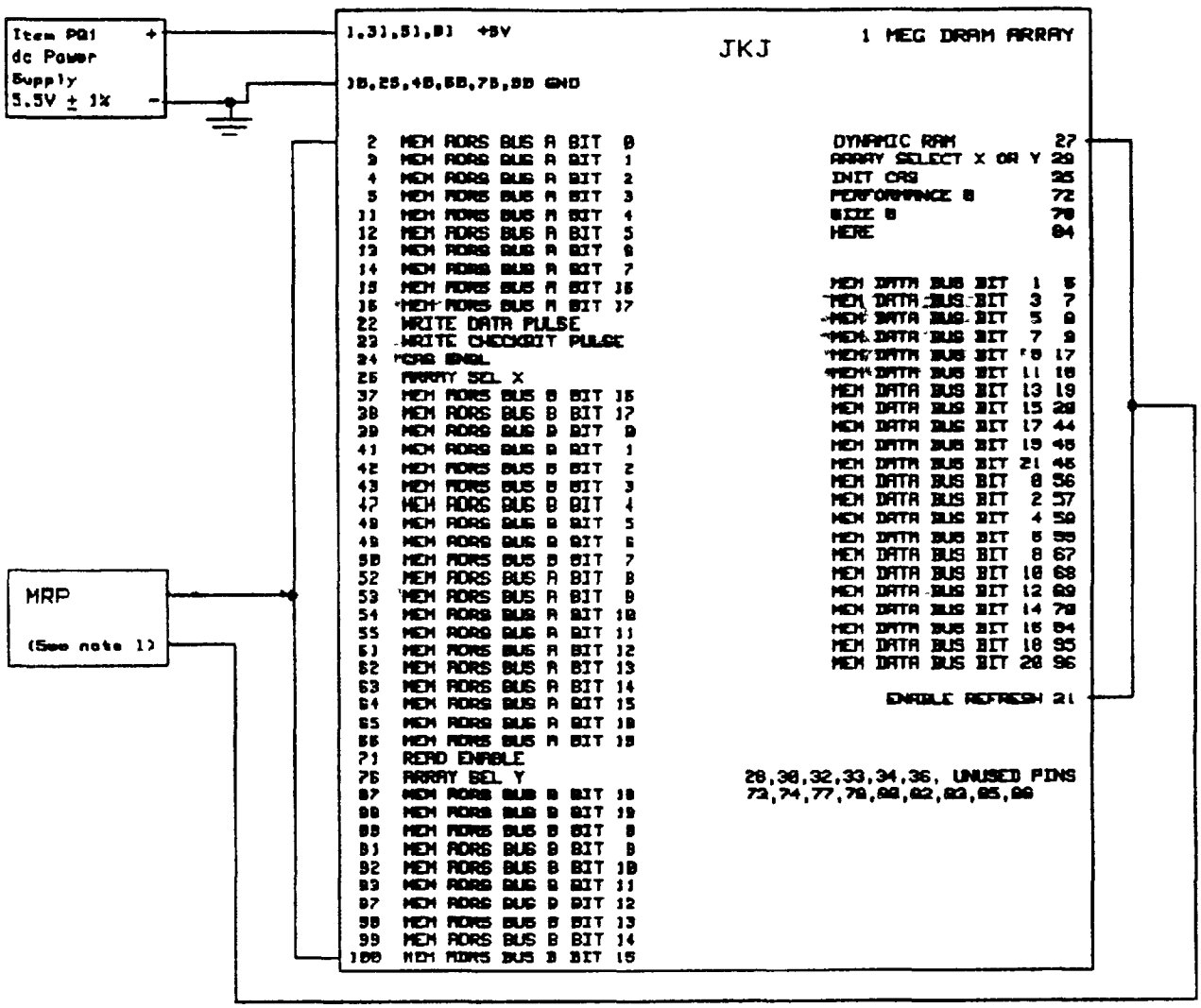
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. JKJ life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 2.53 W (max); 1.693 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MEM ABUS A0	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS A1	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS A2	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS A3	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS A4	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS A5	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS A6	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS A7	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS A16	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS A17	-1.2	0.4	2.4	-0.6	0.02	0.1	50
WR DPULSE	-1.2	0.4	2.4	-4.8	0.06	0.3	50
WR CHK PULSE	-1.2	0.4	2.4	-1.6	0.02	0.1	50
CAS ENBL	-1.2	0.4	2.4	-6.4	0.08	0.4	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
ARRAY SEL X	-1.2	0.4	2.4	-8.0	0.1	0.5	75
MEM ABUS B16	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS B17	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS B0	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS B1	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS B2	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS B3	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS B4	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS B5	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS B6	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS B7	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS A8	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS A9	-1.2	0.4	2.4	-0.6	0.02	0.1	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MEM ABUS A10	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS A11	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS A12	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS A13	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS A14	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS A15	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS A18	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS A19	-1.2	0.4	2.4	-0.6	0.02	0.1	50
READ ENABLE	-1.2	0.4	2.4	-0.03	0.03	0.03	175
ARRAY SEL Y	NA	0.4	2.4	-0.6	0.02	0.1	75
MEM ABUS B18	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS B19	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS B8	-1.2	0.4	2.4	-0.6	0.02	0.1	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MEM ABUS B9	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS B10	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS B11	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS B12	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS B13	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS B14	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM ABUS B15	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM DBUS 1	NA	0.4	2.4	-0.015	0.015	NA	50
MEM DBUS 3	NA	0.4	2.4	-0.015	0.015	NA	50
MEM DBUS 5	NA	0.4	2.4	-0.015	0.015	NA	50
MEM DBUS 7	NA	0.4	2.4	-0.015	0.015	NA	50
MEM DBUS 9	NA	0.4	2.4	-0.015	0.015	NA	50
MEM DBUS 11	NA	0.4	2.4	-0.015	0.015	NA	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MEM DBUS 13	NA	0.4	2.4	-0.015	0.015	NA	50
MEM DBUS 15	NA	0.4	2.4	-0.015	0.015	NA	50
MEM DBUS 17	NA	0.4	2.4	-0.015	0.015	NA	50
MEM DBUS 19	NA	0.4	2.4	-0.015	0.015	NA	50
MEM DBUS 21	NA	0.4	2.4	-0.015	0.015	NA	50
MEM DBUS 0	NA	0.4	2.4	-0.015	0.015	NA	50
MEM DBUS 2	NA	0.4	2.4	-0.015	0.015	NA	50
MEM DBUS 4	NA	0.4	2.4	-0.015	0.015	NA	50
MEM DBUS 6	NA	0.4	2.4	-0.015	0.015	NA	50
MEM DBUS 8	NA	0.4	2.4	-0.015	0.015	NA	50
MEM DBUS 10	NA	0.4	2.4	-0.015	0.015	NA	50
MEM DBUS 12	NA	0.4	2.4	-0.015	0.015	NA	50
MEM DBUS 14	NA	0.4	2.4	-0.015	0.015	NA	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MEM DBUS 16	NA	0.4	2.4	-0.015	0.015	NA	50
MEM DBUS 18	NA	0.4	2.4	-0.015	0.015	NA	50
MEM DBUS 20	NA	0.4	2.4	-0.015	0.015	NA	50

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
MEM DBUS 1	2.4	0.5	-0.4	6.0	NA	NA	-60.0	50
MEM DBUS 3	2.4	0.5	-0.4	6.0	NA	NA	-60.0	50
MEM DBUS 5	2.4	0.5	-0.4	6.0	NA	NA	-60.0	50
MEM DBUS 7	2.4	0.5	-0.4	6.0	NA	NA	-60.0	50
MEM DBUS 9	2.4	0.5	-0.4	6.0	NA	NA	-60.0	50
MEM DBUS 11	2.4	0.5	-0.4	6.0	NA	NA	-60.0	50
MEM DBUS 13	2.4	0.5	-0.4	6.0	NA	NA	-60.0	50
MEM DBUS 15	2.4	0.5	-0.4	6.0	NA	NA	-60.0	50
MEM DBUS 17	2.4	0.5	-0.4	6.0	NA	NA	-60.0	50
MEM DBUS 19	2.4	0.5	-0.4	6.0	NA	NA	-60.0	50
MEM DBUS 21	2.4	0.5	-0.4	6.0	NA	NA	-60.0	50
MEM DBUS 0	2.4	0.5	-0.4	6.0	NA	NA	-60.0	50
MEM DBUS 2	2.4	0.5	-0.4	6.0	NA	NA	-60.0	50

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
MEM DBUS 4	2.4	0.5	-0.4	6.0	NA	NA	-60.0	50
MEM DBUS 6	2.4	0.5	-0.4	6.0	NA	NA	-60.0	50
MEM DBUS 8	2.4	0.5	-0.4	6.0	NA	NA	-60.0	50
MEM DBUS 10	2.4	0.5	-0.4	6.0	NA	NA	-60.0	50
MEM DBUS 12	2.4	0.5	-0.4	6.0	NA	NA	-60.0	50
MEM DBUS 14	2.4	0.5	-0.4	6.0	NA	NA	-60.0	50
MEM DBUS 16	2.4	0.5	-0.4	6.0	NA	NA	-60.0	50
MEM DBUS 18	2.4	0.5	-0.4	6.0	NA	NA	-60.0	50
MEM DBUS 20	2.4	0.5	-0.4	6.0	NA	NA	-60.0	50
DYNAMIC RAM	NA	0.5	-0.4	6.0	0.05	0.05	NA	NA
ARRAYSEL X/Y	2.4	0.5	-0.4	6.0	NA	NA	NA	NA
INIT CAS	2.4	0.5	-0.4	6.0	NA	NA	NA	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
PERFORMANCE0	NA	0.5	-0.4	6.0	0.05	0.05	NA	NA
SIZE 0	NA	0.5	-0.4	6.0	0.05	0.05	NA	NA
HERE	NA	0.5	-0.4	6.0	0.05	0.05	NA	NA

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tsu	MEM ABUS A0-19 to ARRAY SEL X	10	10	10	-	-	-	nS	12
1	tsu	MEM ABUS A0-19 to ARRAY SEL Y	10	10	10	-	-	-	nS	12
1	tsu	MEM ABUS B0-19 to ARRAY SEL X	10	10	10	-	-	-	nS	12
1	tsu	MEM ABUS B0-19 to ARRAY SEL Y	10	10	10	-	-	-	nS	12
2	tsu	ARRAY SEL X to CAS ENBL	27	27	27	-	-	-	nS	12
2	tsu	ARRAY SEL Y to CAS ENBL	27	27	27	-	-	-	nS	12
3	tsu	ARRAY SEL X to READ ENABLE	85	85	85	-	-	-	nS	12
3	tsu	ARRAY SEL Y to READ ENABLE	85	85	85	-	-	-	nS	12
4	tpd	ARRAY SEL X to MEM DBUS 0-21	-	-	-	180	180	180	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	ARRAY SEL Y to MEM DBUS 0-21	-	-	-	180	180	180	nS	12
5	tsu	READ ENABLE to CAS ENBL	1	1	1	-	-	-	nS	12
6	tsu	READ ENABLE to MEM ABUS A0-19	0	0	0	-	-	-	nS	12
6	tsu	READ ENABLE to MEM ABUS B0-19	0	0	0	-	-	-	nS	12
7	tpw	ARRAY SEL X	156	156	156	-	-	-	nS	12
7	tpw	ARRAY SEL Y	156	156	156	-	-	-	nS	12
8	tsu	ARRAY SEL X to WR DPULSE	83	83	83	-	-	-	nS	12
8	tsu	ARRAY SEL Y to WR DPULSE	83	83	83	-	-	-	nS	12
9	tsu	MEM DBUS 0-21 to WR DPULSE	9	9	9	-	-	-	nS	12
10	tpw	WR DPULSE	50	50	50	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
11	tsu	WR DPULSE to MEM ABUS A0-19	8	8	8	-	-	-	nS	12
11	tsu	WR DPULSE to MEM ABUS B0-19	8	8	8	-	-	-	nS	12

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6.4 Module function. The 1M word array contains 1 million x 22-bits of Dynamic Random Access Memory (DRAM), address A and B multiplexers, Row Address Select (RAS), Column Address Select (CAS), control and write drivers, even/odd data buffers, and timing circuits.

6.4.1 DRAM array. The 1M word DRAM array is divided into two subarrays, one for the lower 11 data bits and one for the upper 11 data bits. Each subarray is made up of eleven 1 million x 1-bit DRAM integrated circuit devices. The A address bits address data bits 11-0. The B address bits address data bits 21-12. Data inputs to and outputs from the array are from the 3-state even/odd data buffers. Address inputs are received from the address A and B buffers.

6.4.2 RAS, CAS control and write drivers. The control drivers are 4-bit buffer circuits with ground strapped inputs. These circuits will provide here, size and array type information when strobed by an active array select X or Y and CAS signal gated to the buffer circuits output enable input. The write logic is a 4-bit buffer circuit which passes commands to write data and write checkbits into the DRAMs. The write data pulse provides the DRAM write enable A, B or C; and, the write checkbit pulse is applied to the upper even and odd data DRAM write enable input. RAS is used to latch the first 10 address bits at the DRAM. CAS latches the second group of address bits.

6.4.3 Address A and B buffers. The address A and B buffers provide DRAM isolation from the memory address bus. Outputs are constantly enabled. The A address bits are applied to the lower data bit DRAMs while the B address bits are applied to the upper data bit DRAMs.

6.4.4 Even/Odd data buffers. The even/odd data buffers are 3-state data buffer devices used to provide isolation between the memory data bus and the DRAMs. Data outputs are enabled during a read of the array.

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6.4.5 Description of I/O pins.

Signal name	Pin number	I/O	Description
MEM ABUS A0	2	I	MEM ADRS BUS A BIT 0
MEM ABUS A1	3	I	MEM ADRS BUS A BIT 1
MEM ABUS A2	4	I	MEM ADRS BUS A BIT 2
MEM ABUS A3	5	I	MEM ADRS BUS A BIT 3
MEM ABUS A4	11	I	MEM ADRS BUS A BIT 4
MEM ABUS A5	12	I	MEM ADRS BUS A BIT 5
MEM ABUS A6	13	I	MEM ADRS BUS A BIT 6
MEM ABUS A7	14	I	MEM ADRS BUS A BIT 7
MEM ABUS A16	15	I	MEM ADRS BUS A BIT 16
MEM ABUS A17	16	I	MEM ADRS BUS A BIT 17
WR DPULSE	22	I	WRITE DATA PULSE
WR CHK PULSE	23	I	WRITE CHECKBIT PULSE
CAS ENBL	24	I	CAS ENBL
ARRAY SEL X	26	I	ARRAY SEL X
MEM ABUS B16	37	I	MEM ADRS BUS B BIT 16
MEM ABUS B17	38	I	MEM ADRS BUS B BIT 17
MEM ABUS B0	39	I	MEM ADRS BUS B BIT 0
MEM ABUS B1	41	I	MEM ADRS BUS B BIT 1
MEM ABUS B2	42	I	MEM ADRS BUS B BIT 2
MEM ABUS B3	43	I	MEM ADRS BUS B BIT 3

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6.4.5 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MEM ABUS B4	47	I	MEM ADRS BUS B BIT 4
MEM ABUS B5	48	I	MEM ADRS BUS B BIT 5
MEM ABUS B6	49	I	MEM ADRS BUS B BIT 6
MEM ABUS B7	50	I	MEM ADRS BUS B BIT 7
MEM ABUS A8	52	I	MEM ADRS BUS A BIT 8
MEM ABUS A9	53	I	MEM ADRS BUS A BIT 9
MEM ABUS A10	54	I	MEM ADRS BUS A BIT 10
MEM ABUS A11	55	I	MEM ADRS BUS A BIT 11
MEM ABUS A12	61	I	MEM ADRS BUS A BIT 12
MEM ABUS A13	62	I	MEM ADRS BUS A BIT 13
MEM ABUS A14	63	I	MEM ADRS BUS A BIT 14
MEM ABUS A15	64	I	MEM ADRS BUS A BIT 15
MEM ABUS A18	65	I	MEM ADRS BUS A BIT 18
MEM ABUS A19	66	I	MEM ADRS BUS A BIT 19
READ ENABLE	71	I	READ ENABLE
ARRAY SEL Y	76	I	ARRAY SEL Y
MEM ABUS B18	87	I	MEM ADRS BUS B BIT 18
MEM ABUS B19	88	I	MEM ADRS BUS B BIT 19
MEM ABUS B8	89	I	MEM ADRS BUS B BIT 8
MEM ABUS B9	91	I	MEM ADRS BUS B BIT 9

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6.4.5 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MEM ABUS B10	92	I	MEM ADRS BUS B BIT 10
MEM ABUS B11	93	I	MEM ADRS BUS B BIT 11
MEM ABUS B12	97	I	MEM ADRS BUS B BIT 12
MEM ABUS B13	98	I	MEM ADRS BUS B BIT 13
MEM ABUS B14	99	I	MEM ADRS BUS B BIT 14
MEM ABUS B15	100	I	MEM ADRS BUS B BIT 15
MEM DBUS 1	6	I/O	MEM DATA BUS BIT 1
MEM DBUS 3	7	I/O	MEM DATA BUS BIT 3
MEM DBUS 5	8	I/O	MEM DATA BUS BIT 5
MEM DBUS 7	9	I/O	MEM DATA BUS BIT 7
MEM DBUS 9	17	I/O	MEM DATA BUS BIT 9
MEM DBUS 11	18	I/O	MEM DATA BUS BIT 11
MEM DBUS 13	19	I/O	MEM DATA BUS BIT 13
MEM DBUS 15	20	I/O	MEM DATA BUS BIT 15
MEM DBUS 17	44	I/O	MEM DATA BUS BIT 17
MEM DBUS 19	45	I/O	MEM DATA BUS BIT 19
MEM DBUS 21	46	I/O	MEM DATA BUS BIT 21
MEM DBUS 0	56	I/O	MEM DATA BUS BIT 0
MEM DBUS 2	57	I/O	MEM DATA BUS BIT 2
MEM DBUS 4	58	I/O	MEM DATA BUS BIT 4

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6.4.5 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MEM DBUS 6	59	I/O	MEM DATA BUS BIT 6
MEM DBUS 8	67	I/O	MEM DATA BUS BIT 8
MEM DBUS 10	68	I/O	MEM DATA BUS BIT 10
MEM DBUS 12	69	I/O	MEM DATA BUS BIT 12
MEM DBUS 14	70	I/O	MEM DATA BUS BIT 14
MEM DBUS 16	94	I/O	MEM DATA BUS BIT 16
MEM DBUS 18	95	I/O	MEM DATA BUS BIT 18
MEM DBUS 20	96	I/O	MEM DATA BUS BIT 20
DYNAMIC RAM	27	O	DYNAMIC RAM
ARRAYSEL X/Y	29	O	ARRAY SELECT X OR Y
INIT CAS	35	O	INIT CAS
PERFORMANCE0	72	O	PERFORMANCE 0
SIZE 0	79	O	SIZE 0
HERE	84	O	HERE

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0080)

Agent: NW

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	C,28, 30,32, 33,34, 36,73, 74,77, 78,80, 82,83, 85,86	Frame	10	INF	MOhm	1-9
A2	Isolation, used to unused pins	C	28,30, 32,33, 34,36, 73,74, 77,78, 80,82, 83,85, 86	10	INF	MOhm	1-9
A3-A16	Isolation, unused to unused pins	see note 4		10	INF	MOhm	1-9

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NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1-A16
Isolation measurement	NA	± 500 KOhm	A1-A16

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. The unused pins on this module are: 28,30,32,33,34,36,73, 74,77,78,80,82,83,85,86.

5. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

6. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

7. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

8. The test circuit shall be as indicated on figure 6.

9. Abbreviations used are defined in 6.1.

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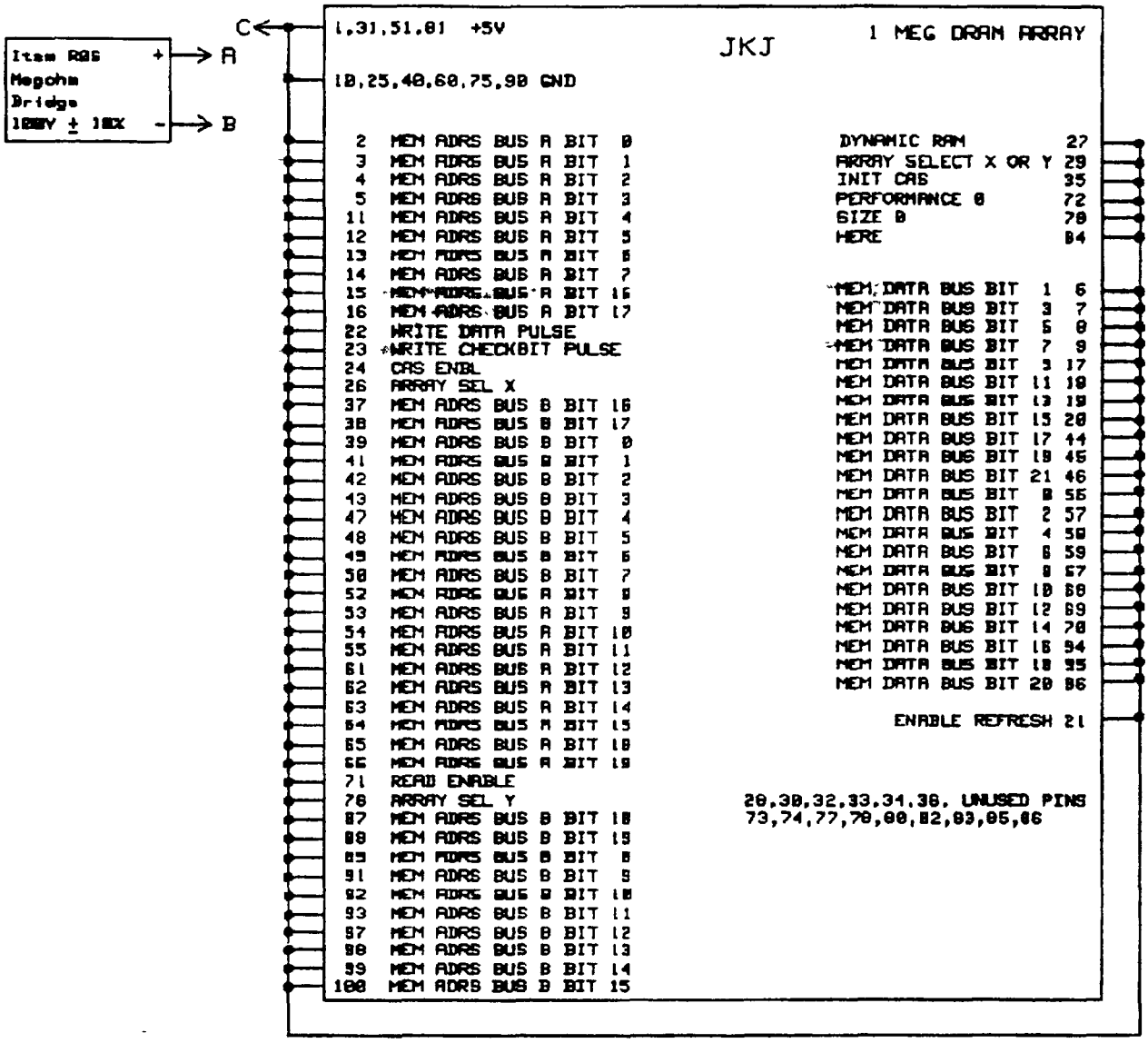


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, GND	10	60				
B6	Continuity, GND	10	21				
B7	Continuity, Vcc	1	31				
B8	Continuity, Vcc	51	81				
B9	Continuity, Vcc	1	51	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B9

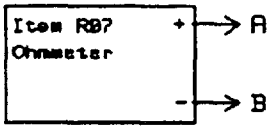
2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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1,31,51,81 +5V

JKJ

1 MEG DRAM ARRAY

10,25,40,50,75,90 GND

2	MEM ADRS BUS A BIT 8			DYNAMIC RAM			27
3	MEM ADRS BUS A BIT 1			ARRAY SELECT X OR Y			29
4	MEM ADRS BUS A BIT 2			INIT CAS			35
5	MEM ADRS BUS A BIT 9			PERFORMANCE 0			72
11	MEM ADRS BUS A BIT 4			SIZE 0			79
12	MEM ADRS BUS A BIT 5			HERE			84
13	MEM ADRS BUS A BIT 6						
14	MEM ADRS BUS A BIT 7						
15	MEM ADRS BUS A BIT 16			MEM DATA BUS BIT 1			6
16	MEM ADRS BUS A BIT 17			MEM DATA BUS BIT 3			7
22	WRITE DATA PULSE			MEM DATA BUS BIT 5			8
23	WRITE CHECKBIT PULSE			MEM DATA BUS BIT 7			9
24	CAS ENBL			MEM DATA BUS BIT 9			17
26	ARRAY SEL X			MEM DATA BUS BIT 11			18
37	MEM ADRS BUS B BIT 16			MEM DATA BUS BIT 13			19
38	MEM ADRS BUS B BIT 17			MEM DATA BUS BIT 15			20
39	MEM ADRS BUS B BIT 0			MEM DATA BUS BIT 17			44
41	MEM ADRS BUS B BIT 1			MEM DATA BUS BIT 19			45
42	MEM ADRS BUS B BIT 2			MEM DATA BUS BIT 21			46
43	MEM ADRS BUS B BIT 3			MEM DATA BUS BIT 0			56
47	MEM ADRS BUS B BIT 4			MEM DATA BUS BIT 2			57
48	MEM ADRS BUS B BIT 5			MEM DATA BUS BIT 4			58
49	MEM ADRS BUS B BIT 6			MEM DATA BUS BIT 6			59
50	MEM ADRS BUS B BIT 7			MEM DATA BUS BIT 8			67
52	MEM ADRS BUS A BIT 8			MEM DATA BUS BIT 10			68
53	MEM ADRS BUS A BIT 9			MEM DATA BUS BIT 12			69
54	MEM ADRS BUS A BIT 10			MEM DATA BUS BIT 14			70
55	MEM ADRS BUS A BIT 11			MEM DATA BUS BIT 16			94
61	MEM ADRS BUS A BIT 12			MEM DATA BUS BIT 18			95
62	MEM ADRS BUS A BIT 13			MEM DATA BUS BIT 20			96
63	MEM ADRS BUS A BIT 14						
64	MEM ADRS BUS A BIT 15			ENABLE REFRESH			21
65	MEM ADRS BUS A BIT 18						
66	MEM ADRS BUS A BIT 19						
71	READ ENABLE						
76	ARRAY SEL Y			28,30,32,33,34,36, UNUSED PINS			
87	MEM ADRS BUS B BIT 18			73,74,77,78,80,82,83,85,86			
88	MEM ADRS BUS B BIT 19						
89	MEM ADRS BUS B BIT 0						
91	MEM ADRS BUS B BIT 9						
92	MEM ADRS BUS B BIT 10						
93	MEM ADRS BUS B BIT 11						
97	MEM ADRS BUS B BIT 12						
98	MEM ADRS BUS B BIT 13						
99	MEM ADRS BUS B BIT 14						
100	MEM ADRS BUS B BIT 15						

FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect		25 ± 5°C Initial limits	Notes
			Positive lead to	Negative lead to		
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	0.828 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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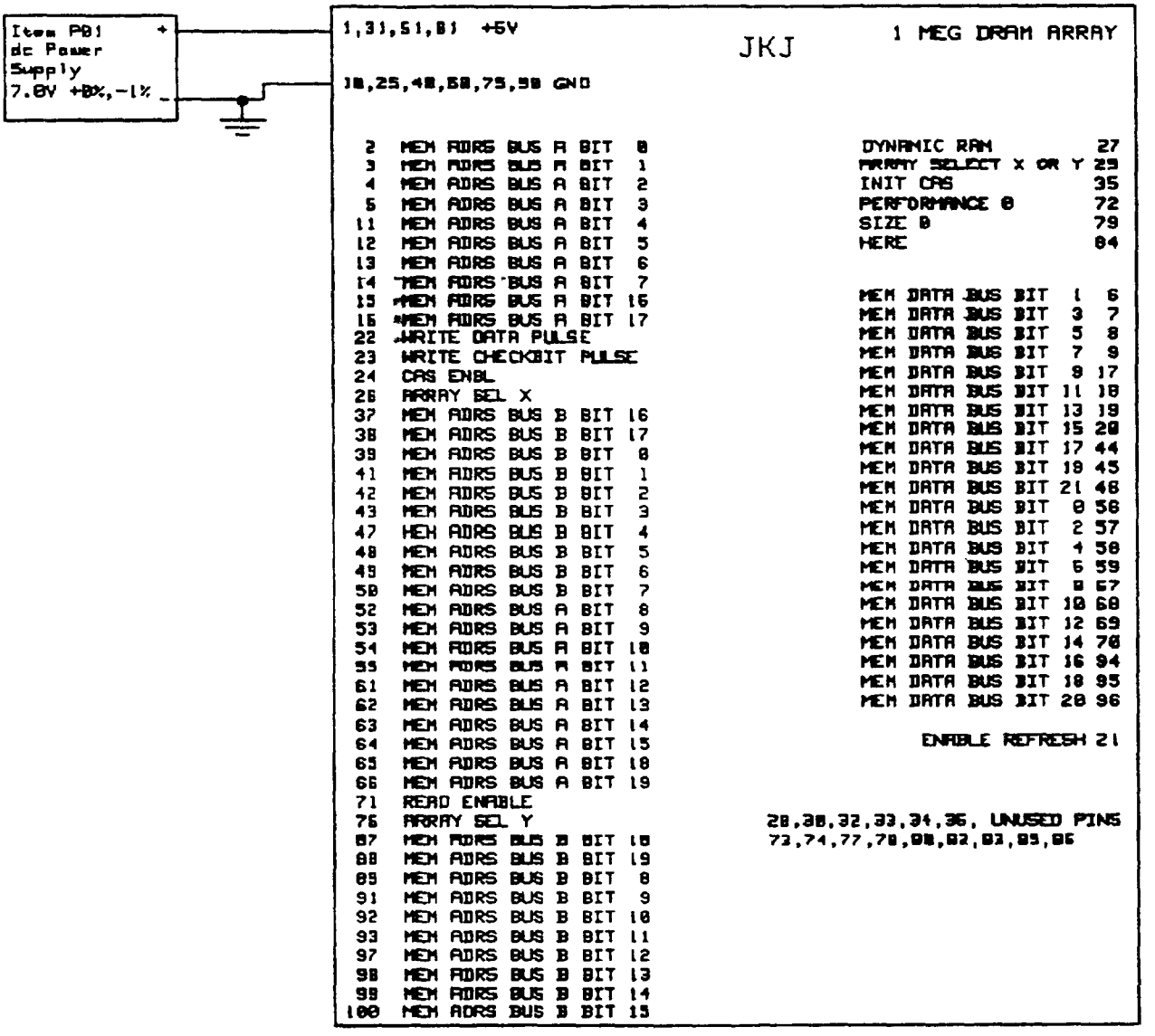


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	MEM ABUS A0	2	5.5	0.0	0.1	mA	1-3
D2	MEM ABUS A1	3					
D3	MEM ABUS A2	4					
D4	MEM ABUS A3	5					
D5	MEM ABUS A4	11					
D6	MEM ABUS A5	12					
D7	MEM ABUS A6	13					
D8	MEM ABUS A7	14					
D9	MEM ABUS A16	15					
D10	MEM ABUS A17	16	5.5	0.0	0.1	mA	1-3
D11	WR DPULSE	22	5.5	0.0	0.3	mA	1-3
D12	WR CHK PULSE	23	5.5	0.0	0.1	mA	1-3
D13	CAS ENBL	24	5.5	0.0	0.4	mA	1-3
D14	ARRAY SEL X	26	5.5	0.0	0.5	mA	1-3
D15	MEM ABUS A8	52	5.5	0.0	0.1	mA	1-3
D16	MEM ABUS A9	53					
D17	MEM ABUS A10	54					
D18	MEM ABUS A11	55					
D19	MEM ABUS A12	61					
D20	MEM ABUS A13	62	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Initial limits Max	Units	Notes
D21	MEM ABUS A14	63	5.5	0.0	0.1	mA	1-3
D22	MEM ABUS A15	64					
D23	MEM ABUS A18	65					
D24	MEM ABUS A19	66	5.5	0.0	0.1	mA	1-3
D25	READ ENABLE	71	5.5	-0.03	0.03	mA	1-3
D26	ARRAY SEL Y	76	5.5	0.0	0.1	mA	1-3
D27	MEM ABUS B16	37					
D28	MEM ABUS B17	38					
D29	MEM ABUS B0	39					
D30	MEM ABUS B1	41					
D31	MEM ABUS B2	42					
D32	MEM ABUS B3	43					
D33	MEM ABUS B4	47					
D34	MEM ABUS B5	48					
D35	MEM ABUS B6	49					
D36	MEM ABUS B7	50					
D37	MEM ABUS B18	87					
D38	MEM ABUS B19	88					
D39	MEM ABUS B8	89					
D40	MEM ABUS B9	91	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D41	MEM ABUS B10	92	5.5	0.0	0.1	mA	1-3
D42	MEM ABUS B11	93					
D43	MEM ABUS B12	97					
D44	MEM ABUS B13	98					
D45	MEM ABUS B14	99					
D46	MEM ABUS B15	100	5.5	0.0	0.1	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D46
Power supply current limit	0.69 A	NA	D1-D46
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D46
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	D1-D46
low, (VL)	0.0 V	± 100 mV	D1-D46
Input current measurement	NA	± 5 uA	D1-D10
	NA	± 15 uA	D11
	NA	± 5 uA	D12
	NA	± 20 uA	D13
	NA	± 25 uA	D14
	NA	± 5 uA	D15-D24
	NA	± 3 uA	D25
	NA	± 5 uA	D26-D46

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	MEM DBUS 1	6	0.0	-350.0	-60.0	mA	1-4
E2	MEM DBUS 3	7					
E3	MEM DBUS 5	8					
E4	MEM DBUS 7	9					
E5	MEM DBUS 9	17					
E6	MEM DBUS 11	18					
E7	MEM DBUS 0	56					
E8	MEM DBUS 2	57					
E9	MEM DBUS 4	58					
E10	MEM DBUS 6	59					
E11	MEM DBUS 8	67					
E12	MEM DBUS 10	68					
E13	MEM DBUS 13	19					
E14	MEM DBUS 15	20					
E15	MEM DBUS 17	44					
E16	MEM DBUS 19	45					
E17	MEM DBUS 21	46					
E18	MEM DBUS 12	69					
E19	MEM DBUS 14	70	0.0	-350.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E20	MEM DBUS 16	94	0.0	-350.0	-60.0	mA	1-4
E21	MEM DBUS 18	95	0.0	-350.0	-60.0	mA	1-4
E22	MEM DBUS 20	96	0.0	-350.0	-60.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E22
Power supply current limit	0.69 A	NA	E1-E22
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E22
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1-E22 E1-E22
Output current measurement	NA	± 14.5 mA	E1-E22

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	MEM ABUS A0	2	2.4	-0.6	0.02	mA	1-3
F2	MEM ABUS A1	3					
F3	MEM ABUS A2	4					
F4	MEM ABUS A3	5	2.4	-0.6	0.02	mA	1-3
F5	MEM DBUS 1	6	2.4	-0.015	0.015	mA	1-3
F6	MEM DBUS 3	7					
F7	MEM DBUS 5	8					
F8	MEM DBUS 7	9	2.4	-0.015	0.015	mA	1-3
F9	MEM ABUS A4	11	2.4	-0.6	0.02	mA	1-3
F10	MEM ABUS A5	12					
F11	MEM ABUS A6	13					
F12	MEM ABUS A7	14					
F13	MEM ABUS A16	15					
F14	MEM ABUS A17	16	2.4	-0.6	0.02	mA	1-3
F15	MEM DBUS 9	17	2.4	-0.015	0.015	mA	1-3
F16	MEM DBUS 11	18					
F17	MEM DBUS 13	19					
F18	MEM DBUS 15	20	2.4	-0.015	0.015	mA	1-3
F19	WR DPULSE	22	2.4	-4.8	0.06	mA	1-3
F20	WR CHK PULSE	23	2.4	-1.6	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	CAS ENBL	24	2.4	-6.4	0.08	mA	1-3
F22	ARRAY SEL X	26	2.4	-8.0	0.1	mA	1-3
F23	MEM DBUS 17	44	2.4	-0.015	0.015	mA	1-3
F24	MEM DBUS 19	45					
F25	MEM DBUS 21	46	2.4	-0.015	0.015	mA	1-3
F26	MEM ABUS A8	52	2.4	-0.6	0.02	mA	1-3
F27	MEM ABUS A9	53					
F28	MEM ABUS A10	54					
F29	MEM ABUS A11	55	2.4	-0.6	0.02	mA	1-3
F30	MEM DBUS 0	56	2.4	-0.015	0.015	mA	1-3
F31	MEM DBUS 2	57					
F32	MEM DBUS 4	58					
F33	MEM DBUS 6	59	2.4	-0.015	0.015	mA	1-3
F34	MEM ABUS A12	61	2.4	-0.6	0.02	mA	1-3
F35	MEM ABUS A13	62					
F36	MEM ABUS A14	63					
F37	MEM ABUS A15	64					
F38	MEM ABUS A18	65					
F39	MEM ABUS A19	66	2.4	-0.6	0.02	mA	1-3
F40	MEM DBUS 8	67	2.4	-0.015	0.015	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F41	MEM DBUS 10	68	2.4	-0.015	0.015	mA	1-3
F42	MEM DBUS 12	69					
F43	MEM DBUS 14	70	2.4	-0.015	0.015	mA	1-3
F44	READ ENABLE	71	2.4	-0.03	0.03	mA	1-3
F45	ARRAY SEL Y	76	2.4	-0.6	0.02	mA	1-3
F46	MEM DBUS 16	94	2.4	-0.015	0.015	mA	1-3
F47	MEM DBUS 18	95					
F48	MEM DBUS 20	96	2.4	-0.015	0.015	mA	1-3
F49	MEM ABUS B16	37	2.4	-0.6	0.02	mA	1-3
F50	MEM ABUS B17	38					
F51	MEM ABUS B0	39					
F52	MEM ABUS B1	41					
F53	MEM ABUS B2	42					
F54	MEM ABUS B3	43					
F55	MEM ABUS B4	47					
F56	MEM ABUS B5	48					
F57	MEM ABUS B6	49					
F58	MEM ABUS B7	50					
F59	MEM ABUS B18	87					
F60	MEM ABUS B19	88	2.4	-0.6	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
F61	MEM ABUS B8	89	2.4	-0.6	0.02	mA	1-3
F62	MEM ABUS B9	91					
F63	MEM ABUS B10	92					
F64	MEM ABUS B11	93					
F65	MEM ABUS B12	97					
F66	MEM ABUS B13	98					
F67	MEM ABUS B14	99					
F68	MEM ABUS B15	100	2.4	-0.6	0.02	mA	1-3

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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F68
Power supply current limit	0.69 A	NA	F1-F68
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F68
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	F1-F68
low, (VL)	0.0 V	± 100 mV	F1-F68
Input current measurement	NA	± 31.0 uA	F1-F4
	NA	± 1.5 uA	F5-F8
	NA	± 31.0 uA	F9-F14
	NA	± 1.5 uA	F15-F18
	NA	± 243.0 uA	F19
	NA	± 81.0 uA	F20
	NA	± 320.0 uA	F21
	NA	± 410.0 uA	F22
	NA	± 1.5 uA	F23-F25
	NA	± 31.0 uA	F26-F29
	NA	± 1.5 uA	F30-F33
	NA	± 31.0 uA	F34-F39
	NA	± 1.5 uA	F40-F43
	NA	± 3.0 uA	F44
	NA	± 31.0 uA	F45
	NA	± 1.5 uA	F46-F48
	NA	± 31.0 uA	F49-F68

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table.

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NOTES: - Continued.

The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	MEM ABUS A0	2	0.4	-0.6	0.02	mA	1-3
G2	MEM ABUS A1	3					
G3	MEM ABUS A2	4					
G4	MEM ABUS A3	5	0.4	-0.6	0.02	mA	1-3
G5	MEM DBUS 1	6	0.4	-0.015	0.015	mA	1-3
G6	MEM DBUS 3	7					
G7	MEM DBUS 5	8					
G8	MEM DBUS 7	9	0.4	-0.015	0.015	mA	1-3
G9	MEM ABUS A4	11	0.4	-0.6	0.02	mA	1-3
G10	MEM ABUS A5	12					
G11	MEM ABUS A6	13					
G12	MEM ABUS A7	14					
G13	MEM ABUS A16	15					
G14	MEM ABUS A17	16	0.4	-0.6	0.02	mA	1-3
G15	MEM DBUS 9	17	0.4	-0.015	0.015	mA	1-3
G16	MEM DBUS 11	18					
G17	MEM DBUS 13	19					
G18	MEM DBUS 15	20	0.4	-0.015	0.015	mA	1-3
G19	WR DPULSE	22	0.4	-4.8	0.06	mA	1-3
G20	WR CHK PULSE	23	0.4	-1.6	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	CAS ENBL	24	0.4	-6.4	0.08	mA	1-3
G22	ARRAY SEL X	26	0.4	-8.0	0.1	mA	1-3
G23	MEM DBUS 17	44	0.4	-0.015	0.015	mA	1-3
G24	MEM DBUS 19	45					
G25	MEM DBUS 21	46	0.4	-0.015	0.015	mA	1-3
G26	MEM ABUS A8	52	0.4	-0.6	0.02	mA	1-3
G27	MEM ABUS A9	53					
G28	MEM ABUS A10	54					
G29	MEM ABUS A11	55	0.4	-0.6	0.02	mA	1-3
G30	MEM DBUS 0	56	0.4	-0.015	0.015	mA	1-3
G31	MEM DBUS 2	57					
G32	MEM DBUS 4	58					
G33	MEM DBUS 6	59	0.4	-0.015	0.015	mA	1-3
G34	MEM ABUS A12	61	0.4	-0.6	0.02	mA	1-3
G35	MEM ABUS A13	62					
G36	MEM ABUS A14	63					
G37	MEM ABUS A15	64					
G38	MEM ABUS A18	65					
G39	MEM ABUS A19	66	0.4	-0.6	0.02	mA	1-3
G40	MEM DBUS 8	67	0.4	-0.015	0.015	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G41	MEM DBUS 10	68	0.4	-0.015	0.015	mA	1-3
G42	MEM DBUS 12	69					
G43	MEM DBUS 14	70	0.4	-0.015	0.015	mA	1-3
G44	READ ENABLE	71	0.4	-0.03	0.03	mA	1-3
G45	ARRAY SEL Y	76	0.4	-0.6	0.02	mA	1-3
G46	MEM DBUS 16	94	0.4	-0.015	0.015	mA	1-3
G47	MEM DBUS 18	95					
G48	MEM DBUS 20	96	0.4	-0.015	0.015	mA	1-3
G49	MEM ABUS B16	37	0.4	-0.6	0.02	mA	1-3
G50	MEM ABUS B17	38					
G51	MEM ABUS B0	39					
G52	MEM ABUS B1	41					
G53	MEM ABUS B2	42					
G54	MEM ABUS B3	43					
G55	MEM ABUS B4	47					
G56	MEM ABUS B5	48					
G57	MEM ABUS B6	49					
G58	MEM ABUS B7	50					
G59	MEM ABUS B18	87					
G60	MEM ABUS B19	88	0.4	-0.6	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G61	MEM ABUS B8	89	0.4	-0.6	0.02	mA	1-3
G62	MEM ABUS B9	91					
G63	MEM ABUS B10	92					
G64	MEM ABUS B11	93					
G65	MEM ABUS B12	97					
G66	MEM ABUS B13	98					
G67	MEM ABUS B14	99					
G68	MEM ABUS B15	100	0.4	-0.6	0.02	mA	1-3

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NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G68
Power supply current limit	0.69 A	NA	G1-G68
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G68
Pattern voltage high, (VH)	3.0 V	± 100 mV	G1-G68
low, (VL)	0.0 V	± 100 mV	G1-G68
Input current measurement	NA	± 31.0 uA	G1-G4
	NA	± 1.5 uA	G5-G8
	NA	± 31.0 uA	G9-G14
	NA	± 1.5 uA	G15-G18
	NA	± 243.0 uA	G19
	NA	± 81.0 uA	G20
	NA	± 320.0 uA	G21
	NA	± 410.0 uA	G22
	NA	± 1.5 uA	G23-G25
	NA	± 31.0 uA	G26-G29
	NA	± 1.5 uA	G30-G33
	NA	± 31.0 uA	G34-G39
	NA	± 1.5 uA	G40-G43
	NA	± 3.0 uA	G44
	NA	± 31.0 uA	G45
	NA	± 1.5 uA	G46-G48
	NA	± 31.0 uA	G49-G68

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table.

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NOTES: - Continued.

The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	MEM DBUS 1	6	-0.4	2.4	4.5	V	1-4
H2	MEM DBUS 3	7					
H3	MEM DBUS 5	8					
H4	MEM DBUS 7	9					
H5	MEM DBUS 9	17					
H6	MEM DBUS 11	18					
H7	ARRAYSEL X/Y	29					
H8	INIT CAS	35					
H9	MEM DBUS 0	56					
H10	MEM DBUS 2	57					
H11	MEM DBUS 4	58					
H12	MEM DBUS 6	59					
H13	MEM DBUS 8	67					
H14	MEM DBUS 10	68					
H15	MEM DBUS 13	19					
H16	MEM DBUS 15	20					
H17	MEM DBUS 17	44					
H18	MEM DBUS 19	45					
H19	MEM DBUS 21	46					
H20	MEM DBUS 12	69	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	MEM DBUS 14	70	-0.4	2.4	4.5	V	1-4
H22	MEM DBUS 16	94					
H23	MEM DBUS 18	95					
H24	MEM DBUS 20	96	-0.4	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H24
Power supply current limit	0.69 A	NA	H1-H24
Forcing current, (Ioh)	-0.4 mA	± 40 uA	H1-H24
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 H1-H24 H1-H24
Output voltage measurement	NA	± 105 mV	H1-H24

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

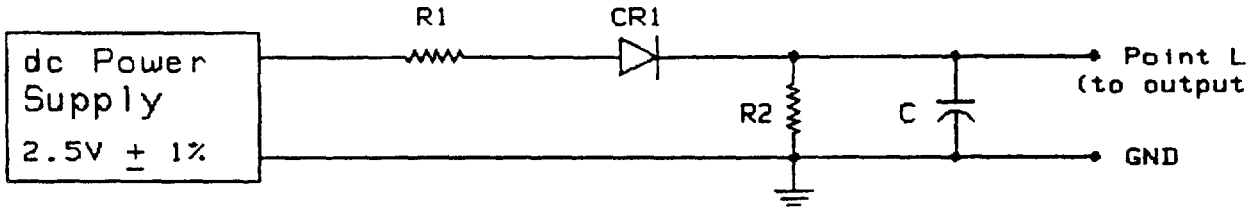
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
6,7,8,9,17,18,19, 20,29,35,44,45, 46,56,57,58,59, 67,68,69,70,94, 95,96	6200	6.0	0.5
27,72,79,84	NA	6.0	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J1	MEM DBUS 1	6	6.0	0.0	0.5	V	1-4
J2	MEM DBUS 3	7					
J3	MEM DBUS 5	8					
J4	MEM DBUS 7	9					
J5	MEM DBUS 9	17					
J6	MEM DBUS 11	18					
J7	DYNAMIC RAM	27					
J8	ARRAYSEL X/Y	29					
J9	INIT CAS	35					
J10	MEM DBUS 0	56					
J11	MEM DBUS 2	57					
J12	MEM DBUS 4	58					
J13	MEM DBUS 6	59					
J14	MEM DBUS 8	67					
J15	MEM DBUS 10	68					
J16	PERFORMANCE0	72					
J17	SIZE 0	79					
J18	HERE	84					
J19	MEM DBUS 13	19					
J20	MEM DBUS 15	20	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J21	MEM DBUS 17	44	6.0	0.0	0.5	V	1-4
J22	MEM DBUS 19	45					
J23	MEM DBUS 21	46					
J24	MEM DBUS 12	69					
J25	MEM DBUS 14	70					
J26	MEM DBUS 16	94					
J27	MEM DBUS 18	95					
J28	MEM DBUS 20	96	6.0	0.0	0.5	V	1-4

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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J28
Power supply current limit	0.69 A	NA	J1-J28
Forcing current, (Iol)	6.0 mA	± 600 uA	J1-J28
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 J1-J28 J1-J28
Output voltage measurement	NA	± 25 mV	J1-J28

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
High level							
K1	DYNAMIC RAM	27	2.4	-0.05	0.05	mA	1-3
K2	PERFORMANCE0	72					
K3	SIZE 0	79					
K4	HERE	84	2.4	-0.05	0.05	mA	1-3
Low level							
K5	DYNAMIC RAM	27	0.4	-0.05	0.05	mA	1-3
K6	PERFORMANCE0	72					
K7	SIZE 0	79					
K8	HERE	84	0.4	-0.05	0.05	mA	1-3

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NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K8
Power supply current limit	0.69 A	NA	K1-K8
Forcing voltage, (Voh) (Vol)	 2.4 V 0.4 V	 ± 50 mV ± 50 mV	 K1-K4 K5-K8
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 K1-K8 K1-K8
Output current measurement	NA	± 5 uA	K1-K8

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L1	MEM ABUS A0	2	-18.0	-1.2	0.0	V	1-3
L2	MEM ABUS A1	3					
L3	MEM ABUS A2	4					
L4	MEM ABUS A3	5					
L5	MEM ABUS A4	11					
L6	MEM ABUS A5	12					
L7	MEM ABUS A6	13					
L8	MEM ABUS A7	14					
L9	MEM ABUS A16	15					
L10	MEM ABUS A17	16					
L11	WR DPULSE	22					
L12	WR CHK PULSE	23					
L13	CAS ENBL	24					
L14	ARRAY SEL X	26					
L15	MEM ABUS B16	37					
L16	MEM ABUS B17	38					
L17	MEM ABUS B0	39					
L18	MEM ABUS B1	41					
L19	MEM ABUS B2	42					
L20	MEM ABUS B3	43	-18.0	-1.2	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L21	MEM ABUS B4	47	-18.0	-1.2	0.0	V	1-3
L22	MEM ABUS B5	48					
L23	MEM ABUS B6	49					
L24	MEM ABUS B7	50					
L25	MEM ABUS A8	52					
L26	MEM ABUS A9	53					
L27	MEM ABUS A10	54					
L28	MEM ABUS A11	55					
L29	MEM ABUS A12	61					
L30	MEM ABUS A13	62					
L31	MEM ABUS A14	63					
L32	MEM ABUS A15	64					
L33	MEM ABUS A18	65					
L34	MEM ABUS A19	66					
L35	READ ENABLE	71					
L36	MEM ABUS B18	87					
L37	MEM ABUS B19	88					
L38	MEM ABUS B8	89					
L39	MEM ABUS B9	91					
L40	MEM ABUS B10	92	-18.0	-1.2	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L41	MEM ABUS B11	93	-18.0	-1.2	0.0	V	1-3
L42	MEM ABUS B12	97	"	"	"	"	"
L43	MEM ABUS B13	98	"	"	"	"	"
L44	MEM ABUS B14	99	"	"	"	"	"
L45	MEM ABUS B15	100	-18.0	-1.2	0.0	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L45
Power supply current limit	NA	NA	L1-L45
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L45
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L45 L1-L45
Input voltage measurement	NA	± 60 mV	L1-L45

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M	PSFC	1	7.57	14.07	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M
Capacitance measurement	NA	± 0.33 uF	M

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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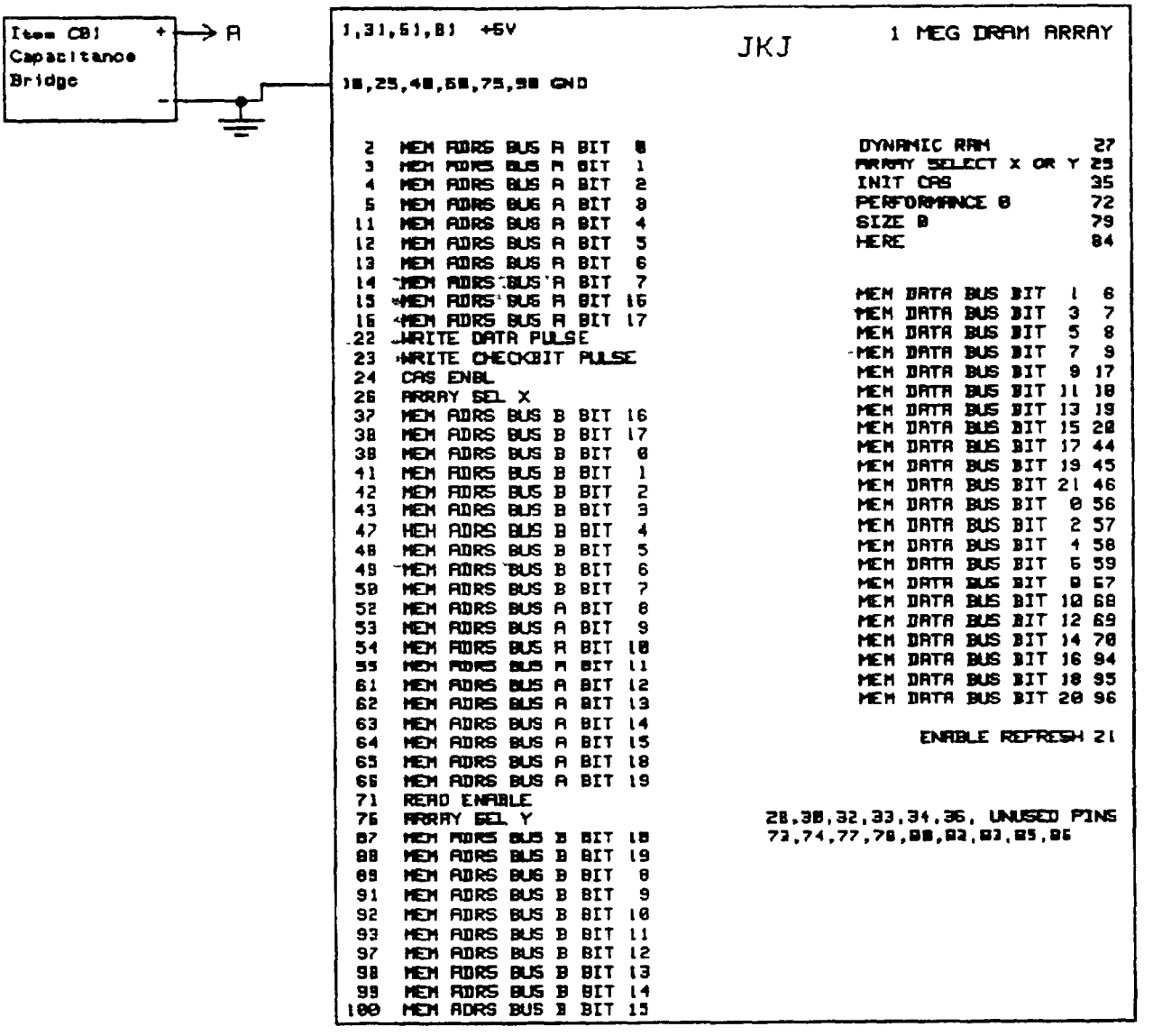


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits		Units	Notes
			Min	Max		
N	Icc	1,31 51,81	190	460	mA	1-3

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NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	0.69 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 13.5 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (225 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (225 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	0.69 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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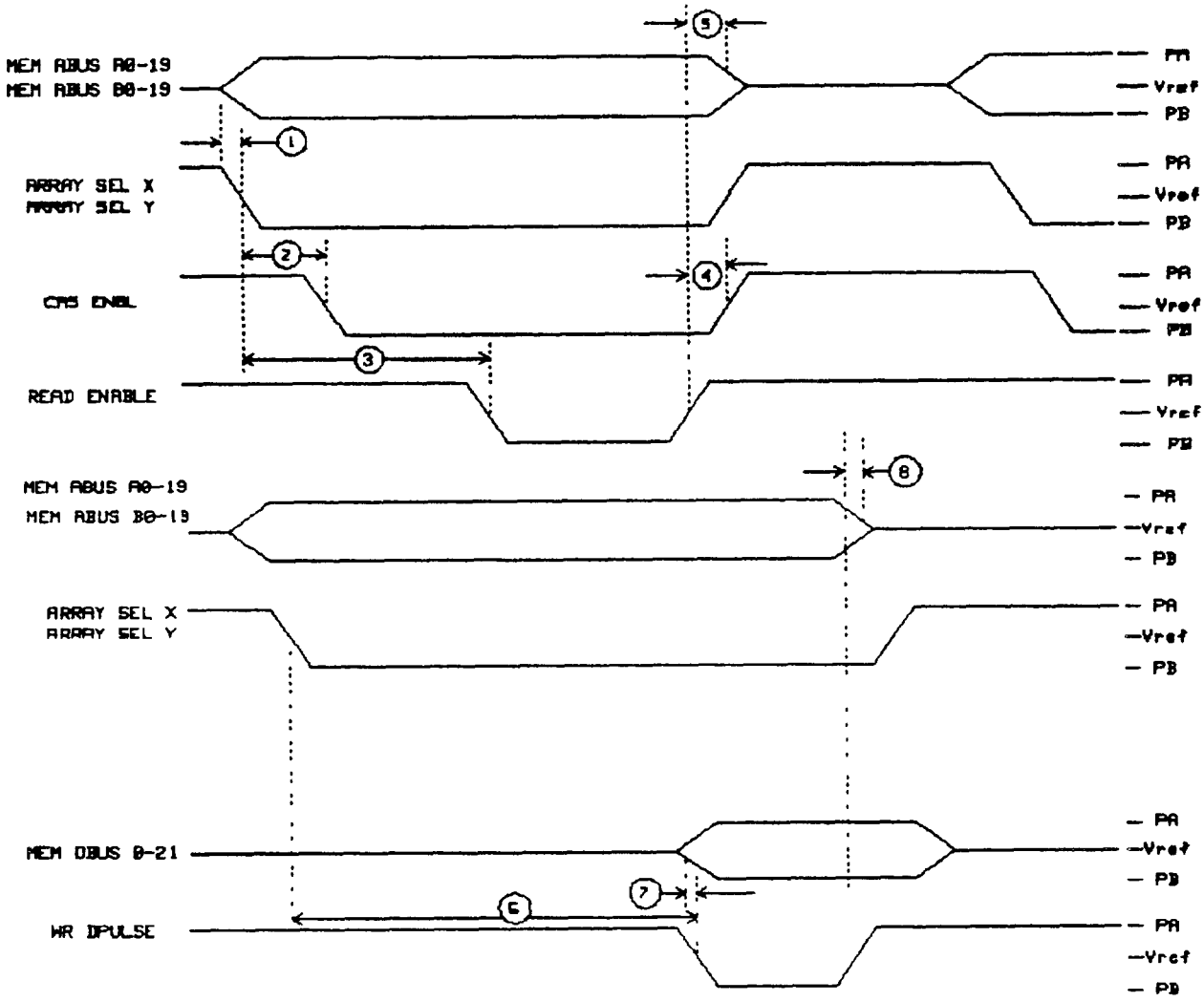


FIGURE 11. Input timing parameters for table XV.

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NOTES:

1. (1) MEM ABUS A0-19/B0-19 to ARRAY SEL X/ARRAY SEL Y.
   Setup time: 10 nS minimum.
2. (2) ARRAY SEL X/ARRAY SEL Y to CAS ENBL.
   Setup time: 27 nS minimum.
3. (3) ARRAY SEL X/ARRAY SEL Y to READ ENABLE.
   Setup time: 85 nS minimum.
4. (4) READ ENABLE to CAS ENBL.
   Setup time: 1 nS minimum.
5. (5) READ ENABLE to MEM ABUS A0-19/B0-19.
   Setup time: 0 nS minimum.
6. (6) ARRAY SEL X/ARRAY SEL Y to WR DPULSE.
   Setup time: 83 nS minimum.
7. (7) MEM DBUS 0-21 to WR DPULSE.
   Setup time: 9 nS minimum.
8. (8) WR DPULSE to MEM ABUS A0-19/B0-19.
   Setup time: 8 nS minimum.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (225 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (225 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (225 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (225 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	0.69 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

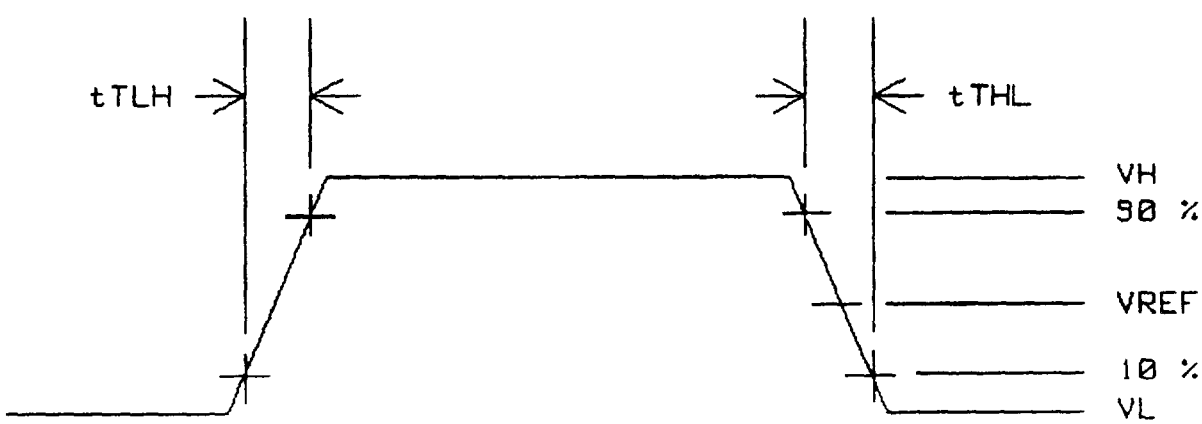
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tsu	MEM ABUS A0-19 to ARRAY SEL X	10	10	10	-	-	-	nS	12
1	tsu	MEM ABUS A0-19 to ARRAY SEL Y	10	10	10	-	-	-	nS	12
1	tsu	MEM ABUS B0-19 to ARRAY SEL X	10	10	10	-	-	-	nS	12
1	tsu	MEM ABUS B0-19 to ARRAY SEL Y	10	10	10	-	-	-	nS	12
2	tsu	ARRAY SEL X to CAS ENBL	27	27	27	-	-	-	nS	12
2	tsu	ARRAY SEL Y to CAS ENBL	27	27	27	-	-	-	nS	12
3	tsu	ARRAY SEL X to READ ENABLE	85	85	85	-	-	-	nS	12
3	tsu	ARRAY SEL Y to READ ENABLE	85	85	85	-	-	-	nS	12
4	tpd	ARRAY SEL X to MEM DBUS 0-21	-	-	-	180	180	180	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	ARRAY SEL Y to MEM DBUS 0-21	-	-	-	180	180	180	nS	12
5	tsu	READ ENABLE to CAS ENBL	1	1	1	-	-	-	nS	12
6	tsu	READ ENABLE to MEM ABUS A0-19	0	0	0	-	-	-	nS	12
6	tsu	READ ENABLE to MEM ABUS B0-19	0	0	0	-	-	-	nS	12
7	tpw	ARRAY SEL X	156	156	156	-	-	-	nS	12
7	tpw	ARRAY SEL Y	156	156	156	-	-	-	nS	12
8	tsu	ARRAY SEL X to WR DPULSE	83	83	83	-	-	-	nS	12
8	tsu	ARRAY SEL Y to WR DPULSE	83	83	83	-	-	-	nS	12
9	tsu	MEM DBUS 0-21 to WR DPULSE	9	9	9	-	-	-	nS	12
10	tpw	WR DPULSE	50	50	50	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
11	tsu	WR DPULSE to MEM ABUS A0-19	8	8	8	-	-	-	nS	12
11	tsu	WR DPULSE to MEM ABUS B0-19	8	8	8	-	-	-	nS	12

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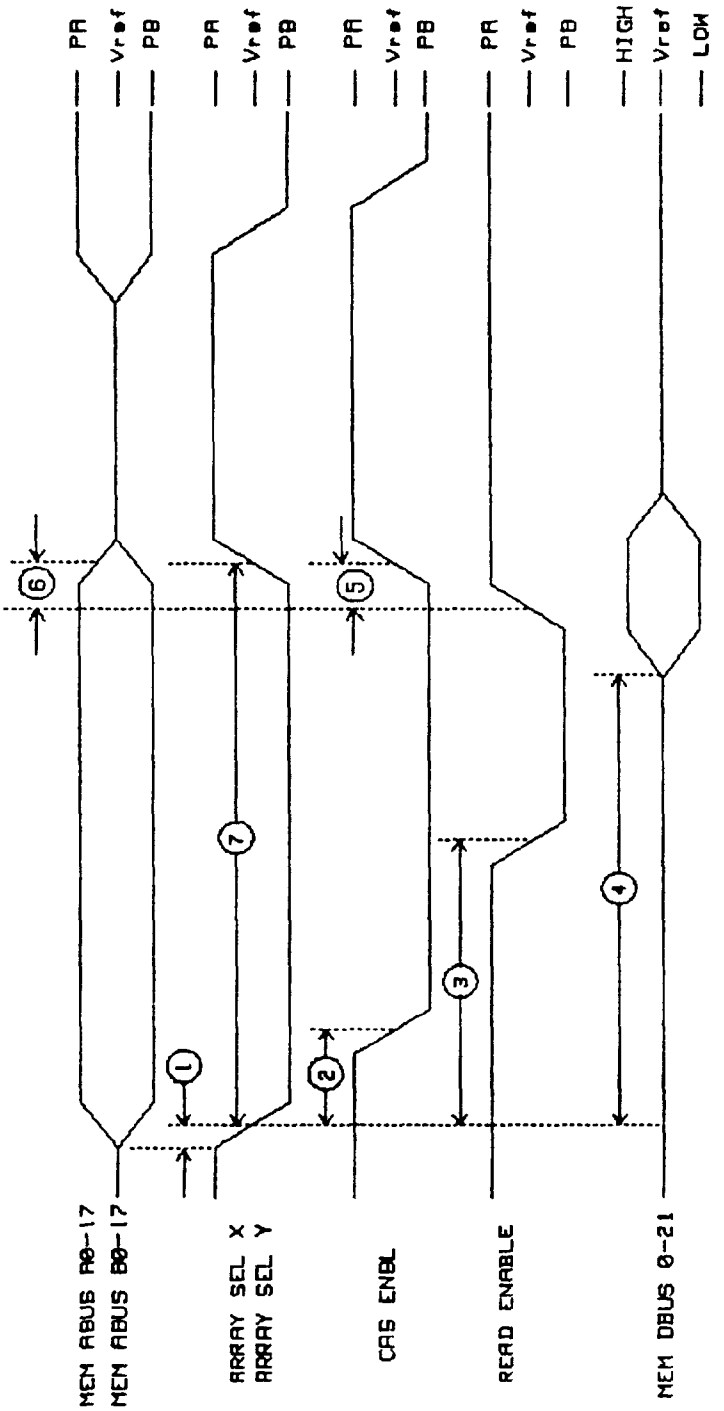


FIGURE 12a. Dynamic waveforms for table XVII.

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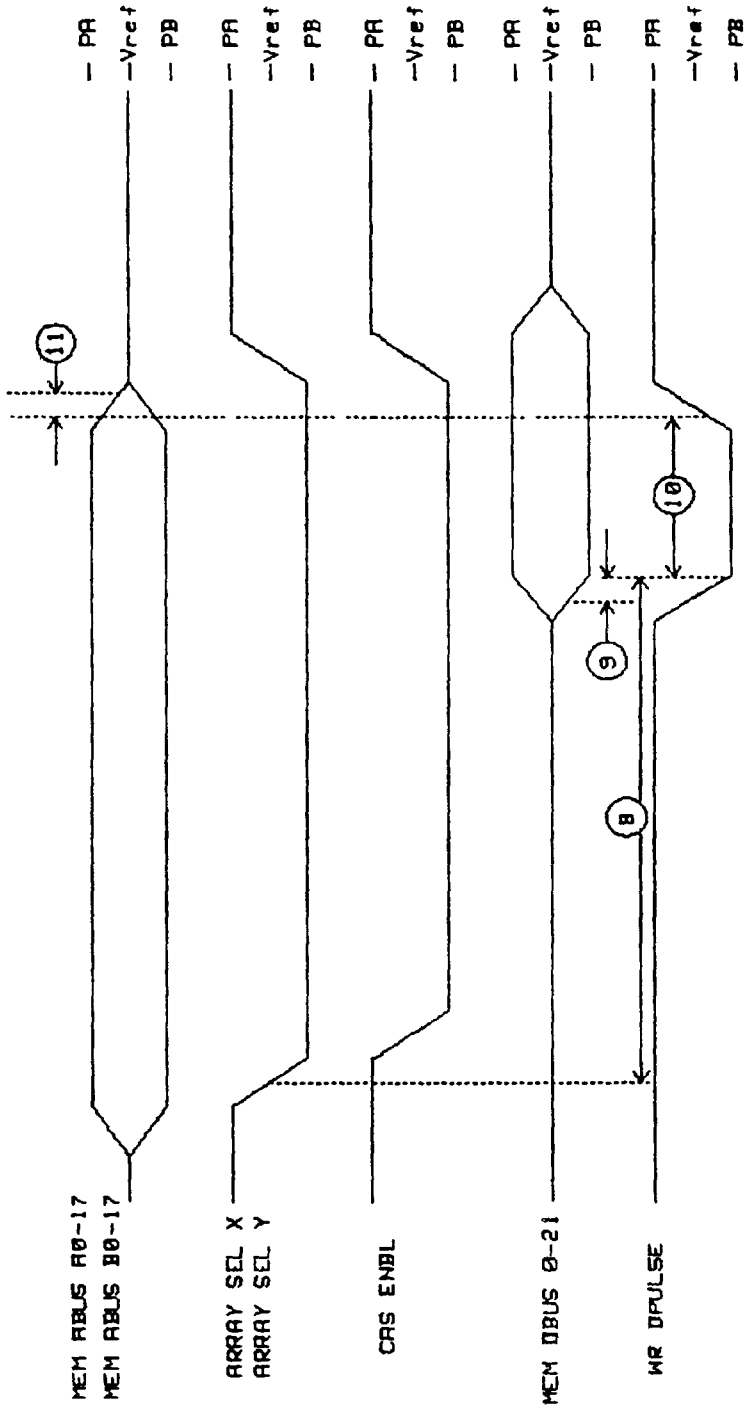


FIGURE 12b. Dynamic waveforms for table XVII.

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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
R1	MEM ABUS A0	2	2	50	pF	1-4
R2	MEM ABUS A1	3				
R3	MEM ABUS A2	4				
R4	MEM ABUS A3	5				
R5	MEM ABUS A4	11				
R6	MEM ABUS A5	12				
R7	MEM ABUS A6	13				
R8	MEM ABUS A7	14				
R9	MEM ABUS A16	15				
R10	MEM ABUS A17	16				
R11	WR DPULSE	22				
R12	WR CHK PULSE	23	2	50	pF	1-4
R13	CAS ENBL	24	2	75	pF	1-4
R14	ARRAY SEL X	26	2	75	pF	1-4
R15	MEM ABUS B16	37	2	50	pF	1-4
R16	MEM ABUS B17	38				
R17	MEM ABUS B0	39				
R18	MEM ABUS B1	41				
R19	MEM ABUS B2	42				
R20	MEM ABUS B3	43	2	50	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	MEM ABUS B4	47	2	50	pF	1-4
R22	MEM ABUS B5	48				
R23	MEM ABUS B6	49				
R24	MEM ABUS B7	50				
R25	MEM ABUS A8	52				
R26	MEM ABUS A9	53				
R27	MEM ABUS A10	54				
R28	MEM ABUS A11	55				
R29	MEM ABUS A12	61				
R30	MEM ABUS A13	62				
R31	MEM ABUS A14	63				
R32	MEM ABUS A15	64				
R33	MEM ABUS A18	65				
R34	MEM ABUS A19	66	2	50	pF	1-4
R35	READ ENABLE	71	2	175	pF	1-4
R36	ARRAY SEL Y	76	2	75	pF	1-4
R37	MEM ABUS B18	87	2	50	pF	1-4
R38	MEM ABUS B19	88				
R39	MEM ABUS B8	89				
R40	MEM ABUS B9	91	2	50	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R41	MEM ABUS B10	92	2	50	pF	1-4
R42	MEM ABUS B11	93				
R43	MEM ABUS B12	97				
R44	MEM ABUS B13	98				
R45	MEM ABUS B14	99				
R46	MEM ABUS B15	100	2	50	pF	1-4
R47	MEM DBUS 1	6	4	50	pF	1-4
R48	MEM DBUS 3	7				
R49	MEM DBUS 5	8				
R50	MEM DBUS 7	9				
R51	MEM DBUS 9	17				
R52	MEM DBUS 11	18				
R53	MEM DBUS 13	19				
R54	MEM DBUS 15	20				
R55	MEM DBUS 17	44				
R56	MEM DBUS 19	45				
R57	MEM DBUS 21	46				
R58	MEM DBUS 0	56				
R59	MEM DBUS 2	57				
R60	MEM DBUS 4	58	4	50	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
R61	MEM DBUS 6	59	4	50	pF	1-4
R62	MEM DBUS 8	67				
R63	MEM DBUS 10	68				
R64	MEM DBUS 12	69				
R65	MEM DBUS 14	70				
R66	MEM DBUS 16	94				
R67	MEM DBUS 18	95				
R68	MEM DBUS 20	96	4	50	pF	1-4

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NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R68
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R68
Capacitance measurement	NA	± 2.4 pF	R1-R12
	NA	± 3.65 pF	R13,R14
	NA	± 2.4 pF	R15-R34
	NA	± 8.65 pF	R35
	NA	± 3.65 pF	R36
	NA	± 2.4 pF	R37-R46
	NA	± 2.3 pF	R47-R68

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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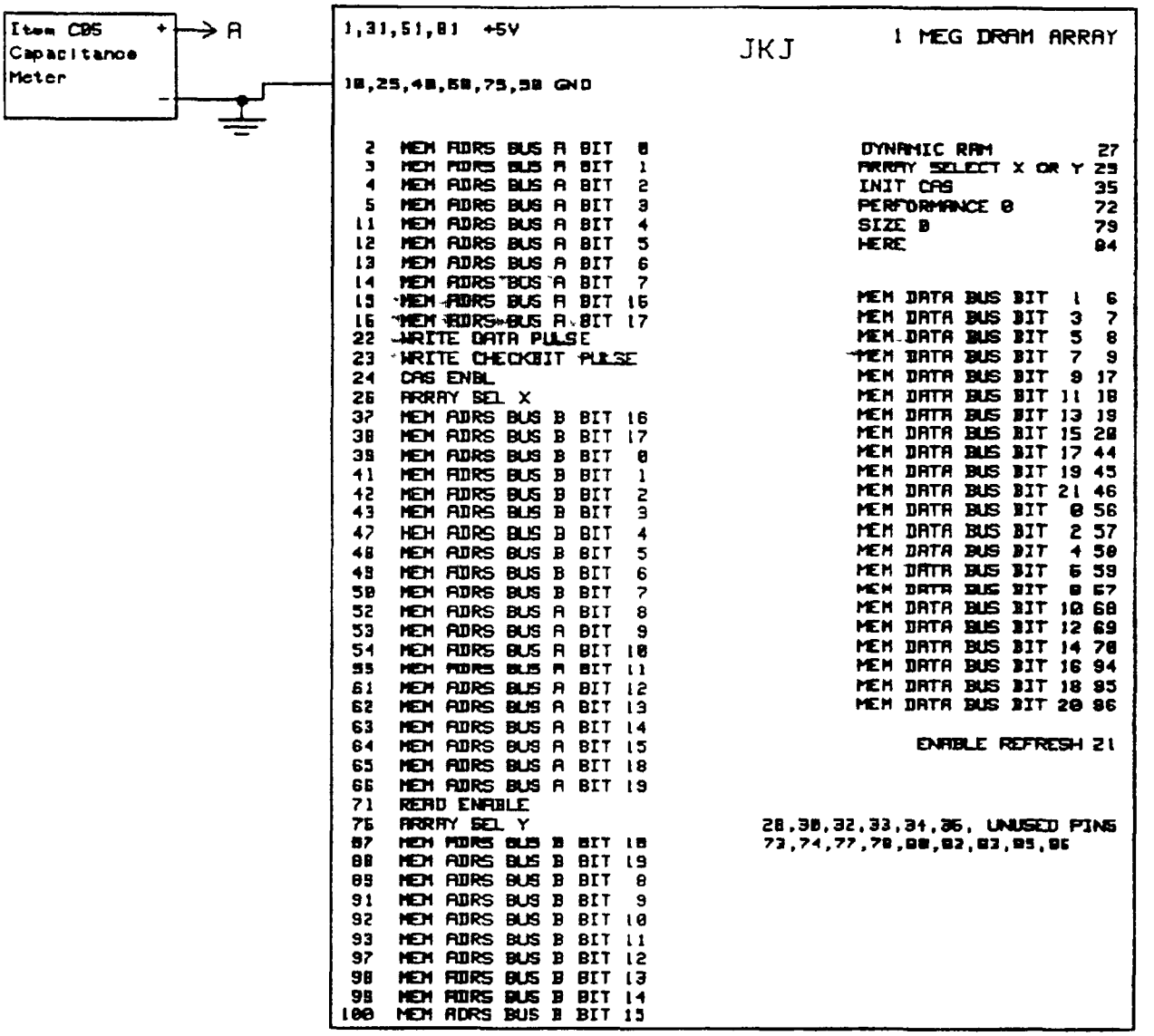


FIGURE 13. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T	PSTA	1	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T
Power supply current limit	0.69 A	NA	T
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T T
Power supply transient amplitude measurement	NA	± 20 mV	T

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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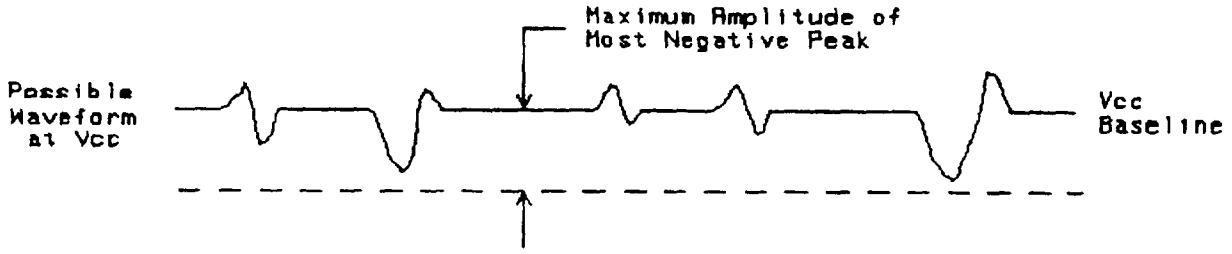
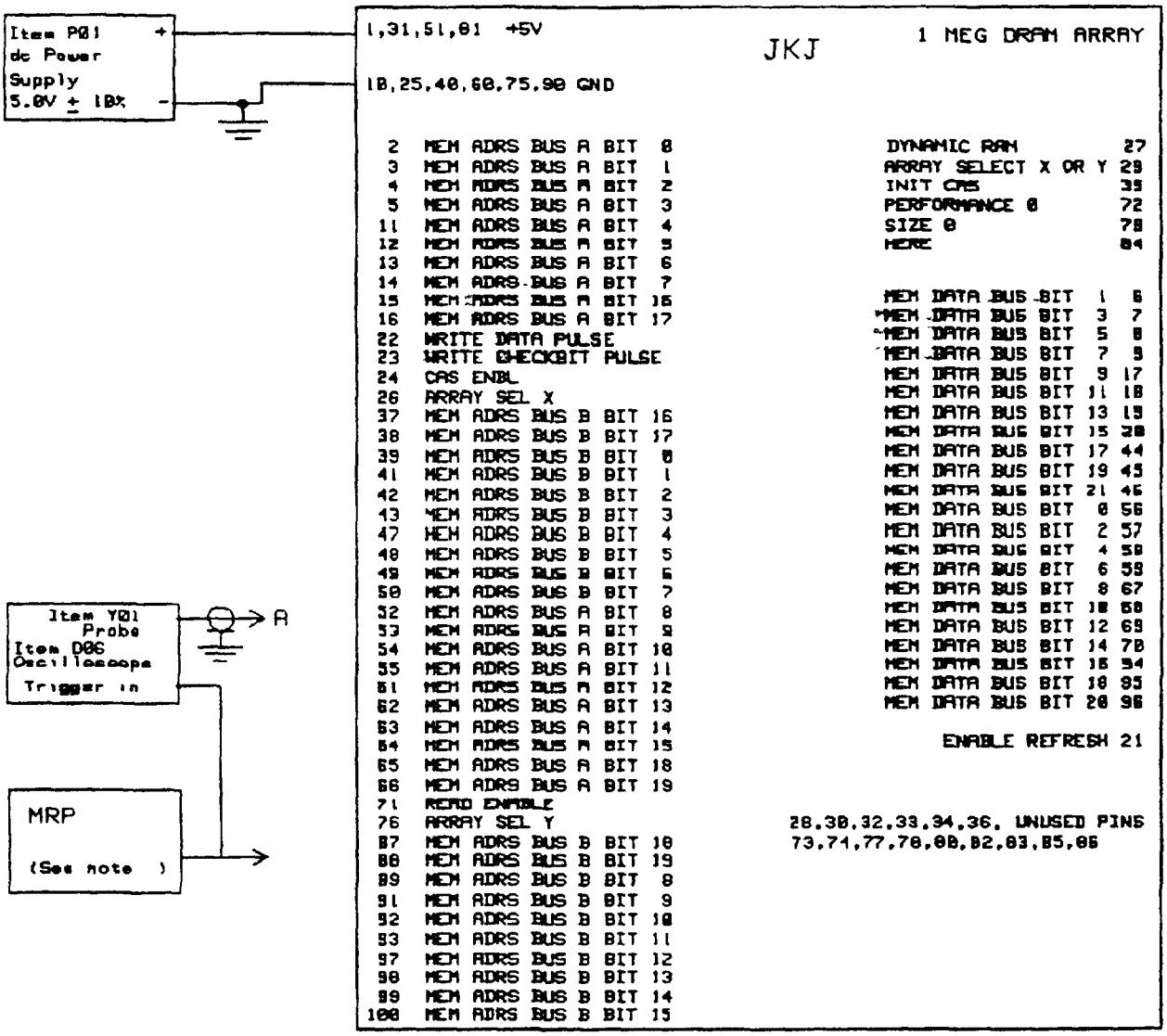


FIGURE 14. Power supply transient amplitude waveform for table XIX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XIX.

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